# Campaign-Shift-Aware Social Bot Detection with Temporal Coordination Modeling and Selective Prediction

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**Abstract**

Social bot detectors are commonly evaluated by randomly splitting accounts, although coordinated accounts may share campaign-specific behavior and therefore violate the practical assumption that deployment data resemble a random sample of the training mixture. This paper develops a campaign-aware evaluation and selective-prediction framework for studying detection reliability under campaign shift. The detector combines account-level profile and activity-rhythm features with a timestamp-aware co-activity graph. Candidate edges connect accounts that act on the same observable object within a temporal neighborhood; a parameterized mixture-of-exponentials kernel weights temporal proximity, while a time-randomization null estimates co-activity expected under a specified reference process. A graph neural network combines individual and relational evidence. For uncertainty, we evaluate bounded campaign-level prediction-set miscoverage using campaign-unit calibration. Because exchangeability across calibration and future campaigns cannot be presumed under campaign shift, the primary analysis treats this quantity empirically; a formal expected-risk guarantee is reported only if its assumptions and finite-sample requirements are independently justified. Neither marginal set coverage nor campaign-level miscoverage implies low error among automatically labeled accounts or robustness to arbitrary distribution shift. Leave-campaign-out cross-fitting is treated as a separate empirical evaluation strategy and is not assumed to inherit a fixed-model conformal guarantee. The experimental protocol compares random, campaign-disjoint, chronological and cross-dataset evaluation; includes graph, temporal and group-aware baselines; and reports detection, coverage, selective utility and computational cost. The manuscript specifies the evaluation protocol; empirical conclusions are reserved for experiments executed on audited datasets.

**Keywords:** Social bot detection; campaign shift; temporal coordination; co-activity graph; conformal risk control; selective prediction

## 1 Introduction

Social bots have become a persistent challenge for online information ecosystems because they can amplify content, manipulate engagement signals, and participate in coordinated information operations. Their detection has consequently evolved from conventional account-level classification based on profile attributes toward approaches that incorporate temporal behavior, social relationships, interaction patterns, and graph-based representations. Machine learning classifiers can exploit account metadata and activity statistics, while graph neural networks (GNNs) enable the integration of relational evidence from interconnected accounts. Temporal representation learning further provides mechanisms for modeling the evolution of interactions and behavioral patterns. Despite these developments, the effectiveness of a social bot detector depends not only on its architecture and input features but also on whether its evaluation protocol reflects the conditions encountered during deployment.

A particularly important challenge arises when the accounts encountered during deployment belong to campaigns that were not represented during model development. Accounts participating in the same campaign may exhibit similar activity rhythms, repeatedly interact with common objects, share profile characteristics, or coordinate their actions within narrow temporal intervals. When accounts are randomly divided into training and test sets, members of the same campaign may appear in both partitions. A detector can consequently exploit campaign-specific regularities that are shared across the partitions, producing an optimistic estimate of its ability to generalize to previously unseen campaigns. This problem is not resolved simply by adopting a more expressive neural architecture: a model may achieve strong random-split performance while remaining sensitive to changes in campaign behavior, coordination strategies, or data-collection conditions.

Campaign-aware evaluation therefore requires a distinction between several forms of generalization. Random account-level evaluation measures performance on held-out accounts drawn from a potentially familiar mixture of campaigns. Campaign-disjoint evaluation instead holds out complete, independently identified campaigns or eligible collection units. Chronological evaluation examines performance on observations occurring after a specified temporal cutoff, whereas cross-dataset evaluation assesses transfer between separately collected corpora. These settings represent different sources of distributional variation and cannot be treated as equivalent evidence of robustness. In particular, chronological separation does not necessarily eliminate campaign overlap, and campaign-disjoint evaluation does not by itself establish generalization across platforms. A rigorous evaluation must therefore measure these settings separately and control information leakage across training, validation, calibration, and test partitions.

Beyond generalization, reliable deployment requires the detector to recognize when its predictions may be insufficiently supported by the available evidence. Conventional classifiers usually produce a class label or probability estimate even for accounts exhibiting unfamiliar behavior or belonging to previously unseen campaigns. Selective prediction provides an alternative by permitting automatic decisions for selected accounts while deferring uncertain cases for additional analysis or human review. However, predictive discrimination, probability calibration, prediction-set coverage, and error among automatically classified accounts are distinct properties. High classification accuracy does not establish reliable uncertainty estimation, and high empirical coverage can be obtained by returning large prediction sets that provide limited decision utility. Moreover, marginal coverage over accounts does not establish low error within every campaign or guarantee reliable decisions under arbitrary distribution shift. Uncertainty assessment must therefore report both reliability and decision utility, with any formal risk-control claim supported by an explicitly defined calibration algorithm and its statistical assumptions.

These challenges motivate the investigation of complementary individual and relational evidence. Account activity rhythms describe the temporal organization of an individual account, but do not necessarily reveal whether its actions are unusually synchronized with those of other accounts. Conversely, a static interaction graph may capture relationships without representing when the associated actions occurred. Temporal co-activity modeling can represent relationships between accounts that act on shared observable objects within a temporal neighborhood. Nevertheless, temporal proximity alone is not sufficient evidence of coordinated automation: popular content, scheduled events, and legitimate communities can generate synchronized activity by chance or in response to common external stimuli. Null-model correction is therefore incorporated to estimate the extent to which observed co-activity exceeds a specified chance-based reference. Its usefulness must be established experimentally, because the validity of the correction depends

on the null model, available observations, and assumptions about the underlying activity process.

This paper investigates a campaign-shift-aware social bot detection framework that integrates account activity rhythms, temporal co-activity relationships, profile characteristics, graph-based prediction, and selective uncertainty assessment. An account-rhythm encoder summarizes individual activity patterns from timestamped events. A temporal co-activity graph represents relationships between accounts based on their actions on shared observable objects, using a multiscale temporal kernel to weight event proximity and null-model correction to discount co-activity expected under a specified reference process. A profile feature extractor represents available account attributes and missingness information. These complementary representations are integrated by a graph neural network to estimate account-level class probabilities. The resulting predictions are evaluated under random, campaign-disjoint, chronological, and cross-dataset settings, while a separate calibration and selective-prediction stage examines prediction-set reliability, automatic-decision coverage, and abstention.

The uncertainty component requires particular methodological care. We formulate campaign-unit calibration in terms of a bounded campaign-level loss and distinguish empirical reliability from formal risk control. A theoretical expected-risk guarantee is meaningful only when the implemented calibration procedure satisfies the assumptions of its associated result, including the definition of the calibration unit, the relevant exchangeability conditions, and the required properties of the loss and threshold-selection procedure. Exchangeability across calibration campaigns does not automatically imply conditional validity for each future campaign or robustness to arbitrary distribution shift. Similarly, cross-fitting or leave-campaign-out evaluation does not inherit a fixed-model conformal guarantee without an appropriate validity argument. Accordingly, the study treats measured campaign-level miscoverage and formal risk-control guarantees as separate forms of evidence.

The research is organized around eight objectives. First, it quantifies the performance difference between random account-level and campaign-disjoint evaluation to determine whether conventional splitting overestimates generalization. Second, it evaluates whether timestamp-aware coordination improves unseen-campaign detection beyond individual-feature, static-graph, and temporal-graph baselines. Third, it isolates the contribution of null-model correction through controlled comparisons of corrected and uncorrected co-activity representations; this is distinct from the second objective, which evaluates the overall contribution of temporal coordination. Fourth, it assesses campaign-unit calibration using empirical miscoverage, deviation from a prespecified target, and uncertainty across independent evaluation units, while restricting theoretical claims to justified assumptions. Fifth, it characterizes the trade-off between automatic decisions and abstention using prediction-set size, singleton rate, abstention rate, selective accuracy, and selective Macro-F1. Sixth, it tests robustness to missing events, timestamp perturbations, low-activity accounts, and synchronized legitimate communities where reliable annotations are available. Seventh, it measures graph-construction, training, and inference costs as the number of accounts and interactions increases. Eighth, it evaluates cross-dataset transfer and the associated changes in detection, calibration, and selective-prediction behavior.

The principal contributions are summarized as follows:

- **Campaign-aware evaluation:** We establish a leakage-controlled evaluation protocol that distinguishes random account-level performance from campaign-disjoint, chronological, and cross-dataset generalization, thereby enabling a more systematic assessment of performance under different forms of distributional variation.

- **Temporal coordination representation:** We develop a representation that integrates individual activity rhythms, timestamp-aware co-activity relationships, null-model correction, and account-profile characteristics. Controlled comparisons and component ablations are designed to quantify the incremental contribution of each component.
- **Campaign-unit uncertainty assessment:** We formulate campaign-level miscoverage using a bounded loss and specify the assumptions relevant to campaign-unit calibration. The evaluation distinguishes empirical reliability from formal risk-control guarantees rather than inferring theoretical validity from observed coverage alone.
- **Selective prediction under campaign shift:** We integrate selective prediction into the evaluation framework to quantify the relationship between prediction-set reliability, automatic-decision coverage, and abstention, including the implications for human review.
- **Systematic experimental assessment:** We define experiments covering campaign-shift degradation, coordination modeling, null-correction ablations, calibration, robustness, computational scalability, and cross-dataset generalization, with statistical analysis conducted at the appropriate evaluation-unit level.

The contribution of this work is not based merely on combining graph neural networks, temporal features, and conformal prediction, since these methodological families have established precedents. Instead, the study examines their interaction within a campaign-aware evaluation framework and tests three distinct questions: whether temporal coordination improves detection on unseen campaigns, whether the improvement persists under controlled component comparisons, and whether selective uncertainty assessment provides useful reliability–coverage trade-offs under explicitly stated assumptions. The strength of these contributions must be established through comparison with the closest existing methods, verified data, controlled experiments, and statistical evidence. No numerical improvement or formal guarantee is inferred solely from the proposed architecture.

The remainder of this paper is organized as follows. Section 2 reviews related work and identifies the research gap. Section 3 summarizes the system architecture, and Section 4 defines the problem and its assumptions. Section 5 describes the proposed method. Section 6 presents the datasets, evaluation protocols, baselines, metrics, and ablation studies. Section 7 summarizes the framework and identifies directions for future work.

## 2 Related Work and Research Gap

Social bot detection has evolved from account-level classification to multimodal representation learning, heterogeneous graph analysis, dynamic network modeling, and coordinated group detection. Despite these advances, predictive performance on conventional benchmark partitions does not necessarily establish generalization to previously unseen campaigns or reliable automated decisions. This section reviews representative studies, classifies their methodological contributions, and positions the proposed campaign-shift-aware framework relative to existing approaches.

**Individual-feature and multimodal detection.** Cresci [1] reviews a decade of social bot detection research and discusses the evolution of detection strategies in response to increasingly sophisticated automated accounts. Feng et al. [2] introduce TwiBot-20, a benchmark containing account profiles, tweets, and network relationships. Their evaluation demonstrates the importance of diverse and sufficiently informative benchmark data for reproducible model comparison. Feng

Table 1: Classification of representative social bot detection studies according to behavioral representation and evaluation capability.

| **Behavioral representation** | **Detection and representation methods** | **Generalization and reliability considerations** |
|---|---|---|
| Individual and multimodal features | Cresci [1]; TwiBot-20 [2]; SATAR [3] | Feature robustness, account diversity, and evaluation beyond conventional account-level partitions. |
| Temporal behavior and coordination | Chavoshi et al. [4]; Khaund et al. [5] | Distinguishing recurring or coordinated activity from generalization to previously unseen campaigns. |
| Heterogeneous graph learning | BotRGCN [6]; RGT [7]; TwiBot-22 [8] | Relational representation quality, benchmark reproducibility, and campaign-disjoint evaluation. |
| Dynamic graph and group interactions | BotDGT [9]; BotSTIP [10] | Temporal network evolution, group interactions, and generalization beyond the original evaluation setting. |
| Conformal prediction and risk control | Angelopoulos and Bates [11]; Angelopoulos et al. [12] | Calibration assumptions, bounded-loss guarantees, prediction-set coverage, and selective utility. |
| **Proposed framework** | **Temporal co-activity, null-model correction, profile features, and relational prediction** | **Campaign-disjoint evaluation, campaign-unit calibration, and selective prediction under explicit assumptions.** |

et al. [3] propose SATAR, a self-supervised account representation learning approach that uses multiple account information sources to improve bot detection. These studies establish the value of multimodal features and learned account representations. However, the use of diverse features alone does not establish generalization to campaigns excluded from training. In comparison, the proposed framework investigates whether explicit temporal coordination provides additional information beyond individual account representations under campaign-disjoint evaluation.

**Temporal behavior and coordinated activity.** Chavoshi et al. [4] develop a system for detecting and archiving bots using temporal activity patterns and correlations between accounts. Their work illustrates how synchronized behavior can provide evidence that is difficult to obtain from isolated account characteristics. Khaund et al. [5] survey social bot coordination during online campaigns and describe methods for analyzing coordinated account behavior. These studies establish that temporal synchronization and coordinated activity are existing research directions. Consequently, the novelty of the proposed framework cannot rest on introducing coordination analysis alone. Instead, it must demonstrate whether timestamp-aware co-activity construction and null-model correction improve detection on campaigns excluded from model development.

**Heterogeneous graph representation learning.** Feng et al. [6] introduce BotRGCN, which combines relational graph convolution with heterogeneous social relationships and multimodal account information. Feng et al. [7] develop relational graph transformer methods that exploit different types of social relations for bot detection. Feng et al. [8] introduce TwiBot-22, a large-

scale graph-based benchmark that supports the systematic comparison of detection methods across multiple datasets. Collectively, these contributions demonstrate the importance of relational information, heterogeneous graph representations, and standardized evaluation. Nevertheless, a graph-based architecture does not automatically establish robustness to campaign shift. The proposed framework therefore requires controlled comparisons against heterogeneous graph baselines to isolate the additional value of timestamp-aware co-activity modeling.

**Dynamic graph and group-interaction modeling.** He et al. [9] propose BotDGT, which models the evolving structure of social networks through dynamic graph transformers. The method integrates structural information from historical graph snapshots with temporal representations of network evolution. Liu and Ma [10] introduce BotSTIP, which uses spatiotemporal group interactions, hypergraph-based representations, and temporal attention to model relationships among social accounts. BotSTIP is particularly relevant because it already addresses temporal and group-level interactions. Accordingly, the proposed framework must be compared directly with such methods rather than claiming novelty from temporal and relational feature integration alone. Its potential distinction lies in the combination of null-corrected temporal co-activity, explicit campaign-disjoint evaluation, and campaign-unit uncertainty assessment. Whether this combination provides an empirical advantage remains to be established experimentally.

**Conformal prediction and uncertainty control.** Angelopoulos and Bates [11] provide a comprehensive introduction to conformal prediction and explain the assumptions under which prediction sets achieve marginal coverage guarantees. Angelopoulos et al. [12] extend conformal calibration to suitable bounded loss functions through conformal risk control. These results provide a foundation for evaluating prediction reliability beyond classification accuracy. However, marginal coverage and expected-loss control are distinct properties, and neither automatically implies conditional reliability for every previously unseen campaign. The proposed framework therefore defines the calibration unit, bounded loss, and exchangeability assumptions explicitly. Campaign-level risk control must be justified for the specified calibration procedure, while account-level coverage, selective accuracy, and abstention are evaluated separately.

**Research gap and positioning.** The literature establishes that multimodal features, temporal behavior, heterogeneous graphs, dynamic network models, and conformal prediction are valuable methodological components. However, these components address different aspects of detection and cannot be assumed to provide campaign-level reliability when combined without further analysis. Three issues motivate the proposed investigation: the need to distinguish random account-level performance from campaign-disjoint generalization; the need to measure the incremental value of temporal coordination against existing dynamic and group-aware models; and the need to evaluate selective prediction under calibration assumptions consistent with the claimed risk guarantee.

The proposed framework investigates these issues through a unified experimental design comprising temporal co-activity representation, relational prediction, campaign-disjoint and chronological evaluation, campaign-unit calibration, and selective decision-making. Its contribution is not the isolated introduction of temporal graphs or conformal prediction, but the explicit formulation and evaluation of their roles in campaign-shift-aware social bot detection. The novelty and practical value of this formulation must be demonstrated through verified comparisons, leakage-controlled experiments, component ablations, and statistical uncertainty analysis.

# 3 System Architecture

The proposed system architecture integrates temporal behavior analysis, coordination-aware graph construction, relational learning, campaign-aware evaluation, and selective prediction to detect social bots under campaign shift. Its purpose is to determine whether temporal coordination improves detection on previously unseen campaigns and whether uncertainty-aware prediction can identify accounts suitable for automatic classification. The overall processing pipeline is illustrated in Fig. 1.

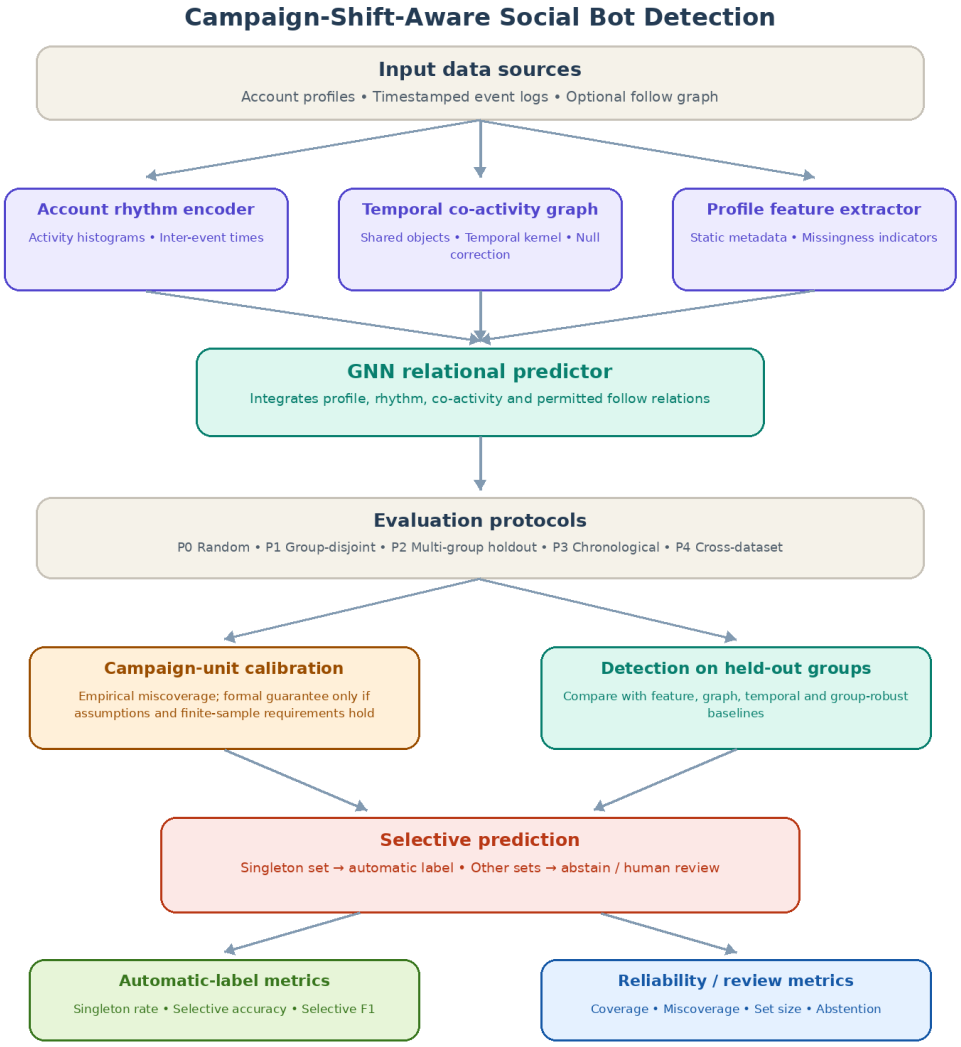



Figure 1: Overview of the campaign-shift-aware social bot detection framework. Calibration and empirical evaluation are separate stages; any risk guarantee depends on the assumptions of the selected calibration procedure.

**Input Data Sources:** The purpose of this component is to provide the information required for bot detection. It collects account profiles, timestamped activity events, optional follow relationships, ground-truth labels, and campaign identifiers. These data provide static account characteristics, temporal behavior, network relationships, and supervision for model training. Campaign identifiers also enable the evaluation of generalization to previously unseen campaigns.

**Account Rhythm Encoder:** This component identifies temporal regularities in individual

account activity. It processes timestamped events to generate weekly activity histograms, inter-event-time distributions, and spectral summaries when sufficient observations are available. These features represent how accounts distribute their activity over time and provide behavioral evidence that complements static profile information. Missingness indicators distinguish insufficient observations from genuine zero-valued measurements.

**Co-Activity Graph Builder:** This component identifies potential coordination between accounts interacting with shared objects at similar times. It computes temporal proximity using a multiscale kernel and constructs weighted relationships between accounts. A time-permutation null model reduces associations attributable to chance temporal overlap under the specified null assumptions. The resulting graph provides relational evidence of unusual co-activity that cannot be captured by analyzing accounts independently.

**Profile Feature Extractor:** This component represents the static characteristics of each account. It processes available metadata, such as account age, follower and following counts, and profile completeness, together with missingness indicators. The resulting representation provides complementary evidence about account characteristics and helps the detector combine profile-based and temporal behavioral information.

**GNN Relational Predictor:** This component combines individual account characteristics with information from related accounts to classify human and bot accounts. It receives profile features, temporal representations, coordination statistics, and the constructed graph. A graph neural network performs message passing over the available co-activity and follow relationships to learn relational account representations. A classification layer converts these representations into class probabilities, enabling the detector to exploit coordination patterns that may be overlooked by feature-only classifiers.

**Evaluation Protocol:** This component determines whether the detector generalizes beyond the accounts and campaigns observed during training. It establishes random account-level, campaign-disjoint, multi-campaign holdout, chronological, and cross-dataset evaluation settings, where applicable. These settings measure different forms of generalization and help reveal performance inflation caused by campaign-specific information leakage. Training, calibration, and final test data are separated to ensure that test labels do not influence model fitting or model selection.

**Campaign-Unit Calibration:** This component selects a decision threshold using held-out calibration units, such as campaigns, without using final test labels. It evaluates the prescribed bounded loss across candidate thresholds and applies the selected conformal risk-control procedure. Its purpose is to provide a statistically principled basis for uncertainty-aware decisions under the assumptions of the corresponding method. If the target risk is infeasible or the required exchangeability assumptions are not justified, this limitation is explicitly reported.

**Detection on Test Campaigns:** This component evaluates the trained detector on previously unseen test accounts. It receives the fixed trained model and the permitted test features and graph information, then produces human and bot class probabilities. The predictions are compared with suitable baseline methods using Macro-F1, precision–recall area under the curve (PR-AUC), and Matthews correlation coefficient (MCC). This component quantifies detection performance under each evaluation protocol without using test labels to adjust the model or calibration threshold.

**Selective Prediction Output:** This component determines whether each account can be classified automatically or requires further review. It receives the prediction set generated by the calibrated decision rule. If the set contains exactly one class, the corresponding label is emitted

automatically. If the set contains multiple classes, the system abstains and refers the account for human review. This mechanism makes the trade-off between automatic classification and uncertainty explicit.

**Output Metrics:** This component measures both classification performance and the reliability of selective decisions. It uses ground-truth test labels, predictions, prediction sets, and abstention decisions to calculate Macro-F1, PR-AUC, MCC, singleton rate, selective accuracy, abstention rate, prediction-set size, empirical coverage, and miscoverage. Reporting these measures separately distinguishes overall discrimination performance from the reliability and coverage of automatic decisions.

The interaction among these components forms an end-to-end pipeline for campaign-aware social bot detection. The account rhythm encoder and profile feature extractor represent individual behavior, while the co-activity graph builder captures temporal relationships. The GNN relational predictor integrates these representations to generate class probabilities. The evaluation protocol establishes the generalization setting, and the calibration and selective prediction components determine which predictions can be issued automatically. Algorithm 1 in Section 5 summarizes the complete processing sequence, including data preparation, model training, calibration, test-time prediction, and performance evaluation.

The detailed processing sequence is specified once in Algorithm 1 in Section 5; this architecture section is intended as a high-level overview rather than a second algorithmic specification.

The complete execution sequence is specified once in Algorithm 1 in Section 5. This section provides the conceptual architecture; the detailed algorithm is not duplicated.

# 4 Problem Formulation

## 4.1 Problem Definition and Data Representation

Let $\mathcal{U}$ denote a collection of observed social media accounts. Each account $u \in \mathcal{U}$ is represented by its profile attributes, temporal activity, and, where available, relational information. Its observable event sequence is defined as

$$\mathcal{E}_u = \{(t_k, a_k, o_k)\}_{k=1}^{n_u}, \tag{1}$$

where $t_k$ denotes the timestamp of event $k$, $a_k$ represents the event or action type, $o_k$ identifies the associated object, and $n_u$ is the number of observed events. Depending on the dataset, an object may correspond to a post, hashtag, or target account. The profile representation is denoted by $\mathbf{p}_u \in \mathbb{R}^{d_p}$, where $d_p$ is the number of available profile features. The binary class label is $y_u \in \{0, 1\}$, with 0 representing a human account and 1 representing a social bot.

When follow relationships are available, they are represented by $\mathcal{G}_f = (\mathcal{U}, \mathcal{E}_f)$, where $\mathcal{E}_f$ denotes the observed follow edges. Follow relationships provide relational context but are not treated as evidence of temporal interaction unless corresponding timestamps are available. All representations are constructed exclusively from fields supported by the selected dataset.

Let $c(u)$ denote the independently annotated campaign membership of account $u$, whenever reliable campaign annotations are available. The set of accounts belonging to campaign $g$ is denoted by $\mathcal{U}_g$. Campaign membership is used to define evaluation groups and calibration units; it is not assumed to be available as a predictive feature at inference time. If campaigns are approximated using collection batches or behavioral heuristics, these groups must be identified as proxies rather than verified campaigns.

The central problem is that a detector evaluated using randomly partitioned accounts may exploit campaign-specific regularities shared across training and test data. Consequently, its measured performance may not represent its ability to identify bots in previously unseen campaigns. The proposed study therefore examines whether temporal coordination modeling improves detection under campaign-disjoint evaluation and whether uncertainty-aware prediction can reduce unreliable automatic decisions.

### 4.2 Campaign-Shift-Aware Generalization

Let $\mathcal{C}_{\text{tr}}$ and $\mathcal{C}_{\text{te}}$ denote the sets of campaigns represented in the training and test partitions, respectively. The campaign-disjoint evaluation protocol imposes

$$\mathcal{C}_{\text{tr}} \cap \mathcal{C}_{\text{te}} = \varnothing. \tag{2}$$

This constraint prevents annotated campaigns from appearing in both partitions. It must be complemented by duplicate removal, identifier-overlap checks, and feature-construction rules that prevent information from the test period or test labels from entering training.

Let $\mathcal{D}_{\text{tr}}$ denote the training distribution and $\mathcal{D}_{\text{te}}$ the test distribution. Campaign shift refers to a change in the distribution of account features, coordination patterns, or class-conditional behavior between training and previously unseen campaigns. In general,

$$\mathcal{D}_{\text{tr}} \neq \mathcal{D}_{\text{te}}, \tag{3}$$

although the precise form and magnitude of this difference must be established empirically. This notation expresses the motivation for campaign-shift-aware evaluation; it does not imply that all distributional changes can be detected or corrected.

Three complementary evaluation settings are considered. First, random account splitting measures performance when accounts may share campaign-specific characteristics across partitions. Second, campaign-disjoint splitting evaluates generalization to unseen campaigns. Third, chronological evaluation restricts feature construction to information available at the prediction time. Cross-dataset evaluation provides an additional test of transfer to a separately collected corpus. These settings are analyzed separately because they represent different sources of distributional variation and are not interchangeable.

### 4.3 Account Representation and Detection

For each account $u$, the proposed detector constructs a representation using three complementary evidence sources: profile characteristics, temporal activity patterns, and temporal coordination with other accounts. Let $\mathbf{h}_u^p$, $\mathbf{h}_u^r$, and $\mathbf{h}_u^c$ denote the corresponding profile, rhythm, and coordination representations. The integrated account representation is defined as

$$\mathbf{x}_u = \phi_\psi \left( \mathbf{h}_u^p \| \mathbf{h}_u^r \| \mathbf{h}_u^c \right), \tag{4}$$

where $\|$ denotes feature concatenation and $\phi_\psi$ denotes the feature-integration function, parameterized separately from the classifier. The representation is subsequently supplied to a classifier parameterized by $\theta$ (with classifier parameters distinct from $\psi$):

$$f_\theta(\mathbf{x}_u) = \widehat{\mathbf{p}}_u = (\widehat{p}_{u,0}, \widehat{p}_{u,1}), \qquad \sum_{y \in \{0,1\}} \widehat{p}_{u,y} = 1, \tag{5}$$

where $\widehat{p}_{u,y}$ is the estimated probability that account $u$ belongs to class $y$.

The conventional class prediction is

$$\widehat{y}_u = \arg\max_{y \in \{0,1\}} \widehat{p}_{u,y}. \tag{6}$$

However, the most probable class is not necessarily sufficiently reliable for automatic deployment. The detector therefore also supports uncertainty-aware prediction, in which a prediction set is constructed and accounts with unresolved predictions may be deferred for human review.

The temporal coordination representation is intended to capture synchronized or closely related activity that cannot be fully described by an account's individual activity rhythm. Its effectiveness is not assumed in advance. It is evaluated through comparisons with profile-only, rhythm-only, and coordination ablations under identical data partitions and evaluation conditions. This design directly tests whether coordination information contributes additional predictive value, particularly when the test campaigns are absent from training.

### 4.4 Prediction Sets and Account-Level Reliability

Let $\Gamma_\tau(\mathbf{x}_u) \subseteq \{0, 1\}$ denote the prediction set returned for account $u$, where $\tau$ is a threshold selected by the specified set-construction and calibration procedure. A singleton set supports an automatic class assignment, whereas a set containing both classes indicates that the system abstains from making an automatic decision. An empty set, if permitted by the chosen construction, also requires review.

The account-level miscoverage indicator is defined as

$$\ell_u(\tau) = \mathbb{I}\left[y_u \notin \Gamma_\tau(\mathbf{x}_u)\right], \tag{7}$$

where $\mathbb{I}[\cdot]$ is the indicator function. If the prediction sets are nested and become smaller as $\tau$ increases, then $\ell_u(\tau)$ is non-decreasing in $\tau$. This monotonicity is conditional on the actual set-construction rule and must be verified for its implementation.

For a campaign $g$ containing $|\mathcal{U}_g|$ evaluated accounts, campaign-level miscoverage is defined as

$$L_g(\tau) = \frac{1}{|\mathcal{U}_g|} \sum_{u \in \mathcal{U}_g} \ell_u(\tau), \qquad 0 \le L_g(\tau) \le 1. \tag{8}$$

This bounded quantity measures the fraction of accounts in a campaign whose true labels are excluded from their prediction sets. Averaging within each campaign gives every campaign a well-defined risk value regardless of its number of accounts. However, the definition alone does not establish statistical exchangeability or a formal risk guarantee.

Let $G_1, \ldots, G_m$ denote the calibration campaigns and $G_{m+1}$ a future campaign. A campaign-unit calibration procedure uses the calibration data to select a threshold $\widehat{\tau}$. For a prespecified risk tolerance $\alpha \in (0, 1)$, the desired expected campaign-level risk target is

$$\mathbb{E}\left[L_{G_{m+1}}(\widehat{\tau})\right] \le \alpha. \tag{9}$$

The expectation is over the relevant calibration and future campaign sampling process. This inequality is a target for evaluation, not an established guarantee of the proposed detector. Its validity depends on the exact calibration algorithm, the definition of the calibration unit, and the assumptions required by the corresponding statistical result. In particular, the necessary

exchangeability conditions must be justified rather than inferred from campaign-disjoint partitioning alone. Arbitrary campaign shift does not automatically satisfy these conditions, and the target does not imply a guarantee for every individual unseen campaign.

Accordingly, campaign-level miscoverage will be reported empirically on held-out campaigns. A formal risk-control claim will be made only if the implemented calibration procedure and its assumptions support the stated result. Otherwise, the analysis will be interpreted as an empirical assessment of campaign-level reliability.

### 4.5 Selective Prediction and Review Burden

For operational use, an account is eligible for automatic labeling when its prediction set contains exactly one class. Define the automatic-label indicator as

$$A_u = \mathbb{I}\left[|\Gamma_\tau(\mathbf{x}_u)| = 1\right]. \tag{10}$$

Accounts whose prediction sets are not singletons are deferred for review. For a test set containing $N$ accounts, the singleton rate and abstention rate are

$$\mathrm{SR} = \frac{1}{N}\sum_{u=1}^{N} A_u, \qquad \mathrm{AR} = 1 - \mathrm{SR}. \tag{11}$$

These measures quantify the fraction of accounts receiving automatic labels and the corresponding review burden.

The correctness of automatic decisions is measured separately using selective accuracy:

$$\mathrm{SA} = \frac{\sum_{u=1}^{N} A_u\, \mathbb{I}[\widehat{y}_u = y_u]}{\sum_{u=1}^{N} A_u}, \tag{12}$$

provided that at least one account receives an automatic label. For singleton prediction sets, $\widehat{y}_u$ is the unique class in $\Gamma_\tau(\mathbf{x}_u)$. If the classifier's highest-probability prediction is not the unique class returned by the prediction-set procedure, the automatic label must instead be defined consistently as that unique class.

Empirical prediction-set coverage is defined as

$$\mathrm{Coverage} = \frac{1}{N}\sum_{u=1}^{N} \mathbb{I}\left[y_u \in \Gamma_\tau(\mathbf{x}_u)\right]. \tag{13}$$

Coverage measures whether the true class is included in the prediction set, whereas selective accuracy measures the correctness of decisions among automatically labeled accounts. A non-singleton set can contain the true class without supporting an automatic decision. Therefore, coverage, campaign-level miscoverage, selective accuracy, singleton rate, and abstention are distinct measures and must not be interpreted as substitutes for one another.

### 4.6 Research Objectives and Evaluation Criteria

The formulation establishes eight linked evaluation objectives.

**O1: Quantify campaign-shift degradation.** Measure the change in detection performance between random account splits and campaign-disjoint splits using Macro-F1, PR-AUC, and complementary classification metrics.

**O2: Evaluate unseen-campaign detection.** Determine whether the proposed representation maintains discriminative performance on campaigns excluded from training.

**O3: Isolate the contribution of temporal coordination.** Compare the complete detector with profile-only, rhythm-only, and coordination-removal variants to quantify the incremental contribution of temporal coordination modeling.

**O4: Assess prediction reliability.** Measure empirical prediction-set coverage and campaign-level miscoverage on held-out campaigns, distinguishing observed performance from any formally justified risk-control guarantee.

**O5: Quantify selective-prediction utility.** Analyze the trade-off between selective accuracy, singleton rate, and abstention as the prediction-set threshold varies.

**O6: Evaluate robustness.** Assess sensitivity to missing events, timestamp perturbations, low-activity accounts, and legitimate synchronized behavior.

**O7: Measure computational cost.** Report training and inference time, memory consumption, and the computational overhead introduced by temporal coordination modeling and calibration.

**O8: Examine cross-dataset generalization.** Evaluate transfer to an independently collected dataset under a documented label-mapping and feature-compatibility protocol.

The evaluation is multi-objective rather than a claim that one scalar score captures all deployment requirements. Computational cost is reported as a tuple

$$\mathbf{C}_{\text{comp}} = (T_{\text{graph}}, T_{\text{train}}, T_{\text{infer}}, M_{\text{peak}}), \tag{14}$$

where the terms denote graph-construction time, model-training time, inference time per account (or batch, stated explicitly), and peak memory consumption. These quantities are reported separately with the hardware and workload specified; they are not combined using arbitrary weights. Detection quality, prediction-set reliability, selective utility, and computational cost are therefore interpreted jointly, because optimizing one criterion need not improve the others.

The experiments are designed to determine whether temporal coordination modeling improves detection on unseen campaigns, whether the resulting predictions remain empirically reliable, and whether selective prediction offers a useful balance between automatic decision quality and review workload. Controlled ablations, leakage-aware data partitions, and separately reported computational measurements provide the evidence needed to assess these objectives.

# 5 Proposed System

The proposed Campaign-Shift-Aware Social Bot Detection framework is designed to identify social bots while evaluating the reliability of detection on previously unseen campaigns. Conventional account-level classification can achieve high performance under random data splits when training and test accounts share campaign-specific behavioral patterns. However, such performance does not necessarily demonstrate generalization to campaigns with different activity rhythms, coordination structures, or account characteristics. The proposed system addresses this limitation by combining individual account representations, temporal coordination modeling, graph-based relational prediction, campaign-aware evaluation, and selective prediction.

As established in the problem formulation, each account $u \in \mathcal{U}$ is represented by a profile vector $\mathbf{p}_u$, a timestamped event sequence $\mathcal{E}_u$, and a binary label $y_u \in \{0, 1\}$, where the two classes denote human and bot accounts. Campaign membership is represented by $c(u)$ whenever

this information is available. The objective is to learn a predictor that exploits both individual behavioral evidence and relationships among accounts while evaluating its performance under explicitly defined campaign and temporal partitions. In addition, the system must distinguish accounts that can be automatically labeled from those for which the available evidence does not support a unique decision.

The proposed system comprises account-level representation learning, temporal co-activity graph construction, null-model correction, relational classification, campaign-aware evaluation, calibration, and selective prediction. The account-level modules characterize individual behavior, whereas the graph module captures temporal relationships between accounts. The relational predictor integrates these sources of evidence, and the final decision modules assess generalization and determine whether a prediction should be issued automatically. The complete execution procedure is summarized in Algorithm 1.

**Input representation and campaign-aware partitioning:** The input component receives account profiles, timestamped events, optional follow relationships, ground-truth labels, and campaign identifiers. Each event is represented as $(t_k, a_k, o_k)$, where $t_k$ denotes the event timestamp, $a_k$ denotes the action type, and $o_k$ denotes the associated observable object. Depending on dataset availability, an object may represent a URL, hashtag, or another shared resource. The optional follow graph is denoted by $\mathcal{G}_f = (\mathcal{U}, \mathcal{E}_f)$.

Before feature construction, the data are partitioned according to the intended evaluation setting. For campaign-disjoint evaluation, training and test campaign sets must satisfy $\mathcal{C}_{\text{tr}} \cap \mathcal{C}_{\text{te}} = \varnothing$. For chronological evaluation, only events available by the prediction cutoff are used to generate features and graph edges. Training, calibration, and final test data are kept separate according to the selected evaluation protocol. The predictor is trained using labeled training accounts, calibration labels are reserved for the prescribed threshold-selection procedure, and test labels are withheld until final evaluation. This separation directly operationalizes the campaign-disjoint and information-leakage constraints specified in the problem formulation.

**Account rhythm representation:** The account rhythm encoder characterizes the temporal organization of individual account activity. Its input is the event sequence $\mathcal{E}_u$ of each account. It extracts complementary statistics that describe activity distribution and temporal regularity. Weekly activity histograms represent how activity is distributed across the week, while inter-event-time distributions characterize the intervals between consecutive events. Spectral summaries can additionally capture periodic patterns when the event history is sufficiently long and informative.

Let $\mathbf{r}_u$ denote the resulting temporal representation. The construction can be expressed as

$$\mathbf{r}_u = \phi_{\text{rhythm}}(\mathcal{E}_u; T), \tag{15}$$

where $\phi_{\text{rhythm}}$ denotes the specified feature-extraction procedure and $T$ denotes the prediction cutoff. The cutoff ensures that no event occurring after the permitted observation time contributes to the representation. Missingness indicators distinguish unavailable or insufficient temporal evidence from genuine zero-valued statistics. Any normalization parameters are estimated from training data and subsequently applied unchanged to calibration and test accounts.

This representation provides individual behavioral evidence but does not, by itself, establish whether an account participates in coordinated activity. The next component therefore models temporal relationships among accounts.

**Temporal co-activity graph construction:** The co-activity graph builder identifies pairs of accounts that interact with shared objects within temporally proximate intervals. For account

$u$, let $\mathcal{E}_u(o)$ denote the timestamps of events associated with object $o$. A multiscale kernel measures temporal proximity at different time scales:

$$\kappa_{\boldsymbol{\theta}}(\delta) = \sum_{m=1}^{M} \pi_m \exp\left(-\frac{\delta}{\lambda_m}\right), \tag{16}$$

subject to $\pi_m \geq 0$, $\sum_{m=1}^{M} \pi_m = 1$, and $\lambda_m > 0$. Here, $\delta$ is the absolute time difference between two events, $\pi_m$ denotes the weight assigned to temporal scale $m$, and $\lambda_m$ controls the decay rate at that scale. Shorter decay parameters emphasize near-simultaneous activity, whereas longer parameters allow associations across wider intervals. The weights and decay parameters determine how evidence from different temporal scales is combined. In this implementation, these parameters are selected using training/validation data and then fixed before final testing; the discrete null estimation and graph sparsification steps are not treated as end-to-end differentiable operations.

The raw co-activity score between accounts $u$ and $v$ is calculated by aggregating temporal proximity over their shared observable objects:

$$\widetilde{w}_{uv} = \sum_{o \in \mathcal{O}_{uv}} \sum_{t \in \mathcal{E}_u(o)} \sum_{t' \in \mathcal{E}_v(o)} \kappa_{\boldsymbol{\theta}}(|t - t'|), \tag{17}$$

where $\mathcal{O}_{uv}$ denotes the set of shared objects observed for the two accounts within the permitted observation window. The score increases when the accounts repeatedly interact with shared objects at temporally proximate times. This operation constructs a candidate relationship between accounts rather than directly assigning a bot label.

The kernel in Equation 16 is a temporal proximity function. It is not a Hawkes-process model and does not establish causal excitation between account activities. Its purpose is to represent multiscale synchronization in a form that can be incorporated into a relational classifier.

**Null-model correction and graph sparsification:** Raw co-activity can be inflated by highly popular objects, common events, or chance temporal overlap. To reduce this effect, the system estimates a null distribution using constrained time randomization within predefined observation intervals. Where the dataset supports it, randomization is stratified by object and coarse time window so that broad object popularity and time-varying activity are not indiscriminately destroyed. The exact randomization scheme is documented and validated against synchronized legitimate activity where suitable annotations are available. The resulting null distribution represents co-activity expected under that specific procedure, not a universal model of organic behavior; a simple within-account permutation alone cannot fully control for shared external events.

Let $\mu^0_{uv}$ and $\sigma^0_{uv}$ denote the estimated mean and standard deviation of the null scores for account pair $(u, v)$. The null-corrected edge weight is defined as

$$w_{uv} = \max\left(0, \frac{\widetilde{w}_{uv} - \mu^0_{uv}}{\sigma^0_{uv} + \varepsilon}\right), \tag{18}$$

where $\varepsilon > 0$ prevents numerical instability when the estimated null standard deviation is small. Pairs with insufficient null variation or too few shared events are handled using a prespecified minimum-support rule and are not allowed to acquire arbitrarily large weights solely because $\sigma^0_{uv}$ is near zero. The corrected score measures the excess of observed co-activity relative to the chosen null distribution. A positive corrected score indicates that the observed value exceeds the

estimated null mean; it does not, on its own, establish statistical significance or malicious intent.

The graph is sparsified by removing edges below a predetermined threshold and retaining at most $k$ neighbors per account. These parameters, the number of permutations, random seeds, active-interval definition, and object-frequency cap are documented and selected using training data only. Where computational approximations are used for popular objects, their effect is validated against more direct permutation estimates on training data. The resulting weighted graph is denoted by $\mathcal{G}_c = (\mathcal{U}, \mathcal{E}_c, \mathbf{W}_c)$, where $\mathcal{E}_c$ contains the retained edges and $\mathbf{W}_c$ contains their corrected weights.

This procedure operationalizes the coordination component of the problem formulation. It reduces the influence of some forms of chance co-activity while preserving relationships that are unusual under the selected null model. Its effectiveness must be established empirically through comparisons with uncorrected and fixed-window graph constructions.

**Coordination statistics and feature integration:** The graph structure provides relational evidence, while account-level coordination statistics summarize the local structure around each account. The system derives measures such as weighted degree, local clustering, and synchronized-event fraction. Let $\mathbf{q}_u$ denote the vector of these coordination statistics. The weighted degree summarizes the strength of an account's retained temporal relationships, local clustering measures connectivity among its neighbors, and synchronized-event fraction characterizes the proportion of its observed activity associated with temporally coordinated events under the chosen definition.

The initial representation combines profile features, temporal features, and coordination statistics:

$$\mathbf{h}_u^{(0)} = \phi_{\text{proj}}\left(\mathbf{p}_u \| \mathbf{r}_u \| \mathbf{q}_u\right), \tag{19}$$

where $\|$ denotes concatenation and $\phi_{\text{proj}}$ denotes a learned projection into a common representation space. The profile vector is processed using the training-fitted transformations, while missingness indicators are retained where necessary. The projection allows heterogeneous features to be integrated into a common input for relational learning. Its architecture and dimensions are specified in the experimental configuration.

This integration is important because static account characteristics, individual activity rhythms, and temporal coordination describe different aspects of behavior. Combining them allows the classifier to use multiple sources of evidence without assuming that any one feature group is sufficient.

**Graph neural network relational prediction:** The relational predictor uses the initial account representations and graph relationships to learn account embeddings for binary classification. Its purpose is to combine individual behavioral evidence with information from connected accounts. If the follow graph is available and permitted by the dataset, it is incorporated as a separate relation type rather than being treated as equivalent to temporal coordination.

Let $\mathcal{R}$ denote the set of available graph relation types and $\mathcal{N}_r(u)$ the neighbors of account $u$ under relation $r$. At message-passing layer $\ell$, the representation is updated as

$$\mathbf{h}_u^{(\ell+1)} = \sigma\left(\mathbf{W}_s^{(\ell)}\mathbf{h}_u^{(\ell)} + \sum_{r\in\mathcal{R}}\sum_{v\in\mathcal{N}_r(u)} \alpha_{uv}^{(r)}\mathbf{W}_r^{(\ell)}\mathbf{h}_v^{(\ell)}\right), \tag{20}$$

where $\mathbf{W}_s^{(\ell)}$ transforms the account's own representation, $\mathbf{W}_r^{(\ell)}$ transforms neighbor representations for relation $r$, $\alpha_{uv}^{(r)}$ is the normalized contribution of neighbor $v$, and $\sigma(\cdot)$ is a nonlinear activation function. The exact neighbor-weighting and normalization rule is fixed in

the implementation and documented in the experimental configuration.

After the final message-passing layer $L$, a classification head produces the human and bot probabilities:

$$\widehat{\mathbf{p}}_u = \mathrm{softmax}\left(\mathbf{W}_{\mathrm{cls}}\mathbf{h}_u^{(L)} + \mathbf{b}_{\mathrm{cls}}\right), \tag{21}$$

where $\mathbf{W}_{\mathrm{cls}}$ and $\mathbf{b}_{\mathrm{cls}}$ are the classifier parameters. The predicted class is the class with the largest estimated probability when a conventional hard prediction is required.

The network is trained using labeled training accounts and a specified classification loss. Model selection is performed without consulting final test labels, and the trained parameters are fixed before calibration. To isolate the contribution of relational modeling, the proposed predictor is compared with a feature-only classifier and a fixed-window graph baseline. Additional ablations remove null correction, temporal rhythm features, coordination statistics, or optional follow relationships. These comparisons determine whether improvements are attributable to the proposed temporal coordination representation rather than simply to the use of a larger model.

**Campaign-aware evaluation and leakage control:** The evaluation component operationalizes the generalization conditions defined in the problem formulation. Random account-level evaluation provides a reference condition, while campaign-disjoint evaluation tests whether the model generalizes to campaigns absent from training. Chronological evaluation restricts feature and edge construction to observations available before the prediction cutoff. Cross-dataset evaluation assesses transfer where compatible labels, attributes, and observation protocols are available.

These settings are reported separately because they measure different forms of generalization. In transductive network evaluation, observable test-account relationships may be used only when they are available under the declared deployment protocol; test labels remain unavailable. In isolated-account evaluation, test-neighbor information is excluded and predictions rely on account-level features. Shared objects across partitions are not automatically leakage, but all cross-partition relationships must follow the declared protocol and be audited. For the primary campaign-disjoint inductive analysis, graph construction and message passing are restricted to the permitted training, calibration, or test partition; cross-partition edges are excluded. Any separate transductive analysis is labeled explicitly and is not used to support an independence-based calibration guarantee unless the required assumptions are established. Preprocessing, hyperparameter selection, graph-construction settings, and model selection are completed without using final test labels.

**Campaign-unit risk calibration:** The calibration component addresses the uncertainty-control objective by selecting a decision parameter using a held-out calibration partition and a prespecified bounded loss. It operates on predictions from the fixed trained model; calibration accounts are not used to update the predictor. When campaigns are selected as calibration units, the method requires an explicit justification for treating those units as exchangeable under the chosen risk-control procedure. Campaign membership alone is insufficient to establish exchangeability.

Let $\Gamma_q(\mathbf{x}_u)$ denote the prediction set for account $u$ at parameter $q$. Using the nonconformity score $s(u, c) = 1 - \widehat{p}_{u,c}$, one possible set construction is

$$\Gamma_q(\mathbf{x}_u) = \{c \in \{0, 1\} : 1 - \widehat{p}_{u,c} \leq q\}. \tag{22}$$

For this construction, increasing $q$ expands the prediction set. Consequently, the account-level

miscoverage loss is non-increasing in $q$. For a calibration unit $g$, the corresponding empirical loss is

$$L_g(q) = \frac{1}{|g|} \sum_{u \in g} \mathbb{I}\left[y_u \notin \Gamma_q(\mathbf{x}_u)\right]. \tag{23}$$

The calibration procedure evaluates this loss over candidate values of $q$ and applies the finite-sample correction specified by the selected conformal risk-control algorithm. Under the theorem's applicable assumptions, the target is to control the marginal expected loss for a future exchangeable unit:

$$\mathbb{E}\left[L_{G_{K+1}}(\widehat{q})\right] \leq \alpha, \tag{24}$$

where $K$ is the number of calibration units, $G_{K+1}$ is a future unit, and $\alpha$ is the prespecified target risk. Equation 24 expresses the intended guarantee under the required assumptions; it is not an unconditional guarantee under arbitrary campaign shift. This loss controls marginal prediction-set miscoverage at the campaign-unit level, not error conditional on receiving an automatic singleton label. Selective accuracy is therefore reported separately, and no selective-risk guarantee is claimed unless a calibration method explicitly controls that loss. The number of independent calibration units and the smallest risk level attainable under the selected finite-sample correction are reported; when too few units are available, the calibration result is treated as underpowered or infeasible. The implementation must specify the exact calibration algorithm, candidate grid, correction, and infeasibility handling. If the target cannot be attained, the result is reported as infeasible rather than silently relaxing the target or adjusting it using test labels.

**Selective prediction and final decision:** After calibration, the selected decision rule is applied unchanged to test accounts. The automatic-decision indicator is defined as

$$A_u = \mathbb{I}\left[|\Gamma_{\widehat{q}}(\mathbf{x}_u)| = 1\right]. \tag{25}$$

When $A_u = 1$, the system emits the unique class in the prediction set as the automatic label. When $A_u = 0$, the system abstains and refers the account for further review. This includes both non-singleton and empty prediction sets; empty sets are counted as abstentions and as miscoverage whenever the true class is excluded. This mechanism distinguishes a unique prediction from an unresolved prediction instead of forcing a label for every account.

The singleton rate measures the proportion of accounts receiving automatic labels:

$$\mathrm{SR} = \frac{1}{N} \sum_{u=1}^{N} A_u, \tag{26}$$

and the abstention rate is

$$\mathrm{AR} = 1 - \mathrm{SR}. \tag{27}$$

Selective accuracy is calculated only among automatically labeled accounts:

$$\mathrm{Acc}_{\mathrm{sel}} = \frac{\sum_{u=1}^{N} A_u \mathbb{I}[\widehat{y}_u = y_u]}{\sum_{u=1}^{N} A_u}, \tag{28}$$

provided that at least one account receives an automatic label. If no account receives an automatic label, selective accuracy is reported as undefined. The singleton rate and abstention

rate must be reported with selective accuracy because a model can increase selective accuracy by abstaining on more difficult accounts.

**Performance assessment:** The final component evaluates both classification performance and uncertainty-aware decision behavior. Macro-F1 measures class-balanced classification quality, PR-AUC evaluates precision–recall discrimination, and MCC summarizes agreement between predicted and ground-truth classes using all confusion-matrix outcomes. The selective-prediction analysis additionally reports singleton rate, abstention rate, selective accuracy, prediction-set size, empirical coverage, and miscoverage.

Empirical coverage is the fraction of test accounts whose true labels are contained in their prediction sets. It is reported separately from selective accuracy and is not presented as a formal guarantee unless the selected calibration theorem and its assumptions justify that interpretation. Similarly, a guarantee on prediction-set miscoverage does not automatically guarantee low error among singleton predictions. If false-discovery risk among automatic labels is an operational requirement, it must be defined and calibrated as a separate objective.

The experimental evaluation compares the complete model against suitable feature-only and graph-based baselines and performs component ablations under the same data-partitioning protocols. Repeated runs, confidence intervals, and appropriate statistical comparisons are used to quantify uncertainty in the reported results. These experiments determine whether the proposed temporal coordination representation improves generalization and whether selective prediction provides a useful trade-off between automatic labeling and abstention.

Algorithm 1 summarizes the complete execution flow. It partitions the data, constructs account representations and the null-corrected temporal graph, trains the relational predictor, and calibrates the decision rule using separate calibration units. The fixed model and calibrated rule are then applied to the final test accounts, after which automatic predictions, abstentions, and evaluation metrics are generated.

# 6 Experimental Evaluation

This section specifies eight named experiments to assess campaign-shift sensitivity, temporal coordination, null-model correction, prediction-set reliability, selective prediction, robustness, computational scalability, and cross-dataset generalization. The experiments are a protocol, not a substitute for executed empirical evaluation. No performance improvement or statistical significance is assumed in advance.

## 6.1 Experimental Setup and Dataset Audit

Candidate datasets include Cresci-2017, TwiBot-20, and TwiBot-22, subject to an audit of data availability and compatibility. Each dataset will be checked for (i) human/bot label definitions; (ii) timestamped events and observable objects; (iii) profile attributes and available relational information; (iv) verified campaign annotations or defensible proxy groups; (v) duplicate accounts, content, identifiers, and potential leakage; and (vi) the number of independent groups, group sizes, and within-group class distributions. Collection batches must not be described as verified operational campaigns unless the source supports that interpretation. When only collection groups or behaviorally inferred groups are available, results will be called group-disjoint or proxy-group-disjoint, as appropriate.

Preprocessing operations, including imputation, scaling, encoding, feature selection, and

**Algorithm 1** Campaign-Shift-Aware Social Bot Detection

**Require:** Account profiles $\mathcal{P}$, events $\mathcal{E}$, labels $\mathcal{Y}$, campaign identifiers $\mathcal{C}$, optional follow graph $\mathcal{G}_f$, cutoff $T$, target risk $\alpha$

**Ensure:** Test predictions, abstention decisions, and evaluation metrics

```
Partition data into training, calibration, and test sets
Enforce campaign separation and temporal cutoffs
Extract profile features and temporal rhythm representations
Fit preprocessing transformations using training data only
Compute multiscale temporal co-activity scores
Estimate the configured time-permutation null distribution
Correct edge weights and sparsify the co-activity graph
Derive coordination statistics and incorporate permitted graph relations
Train the relational predictor using labeled training data
Fix model parameters before calibration
Generate calibration predictions and compute unit-level losses
Apply the selected risk-control algorithm to obtain q̂ or report infeasibility
Generate test predictions using the fixed predictor
Construct test prediction sets using q̂
for all test accounts u do
    if |Γ_q̂(x_u)| = 1 then
        Emit the unique class as an automatic prediction
    else
        Abstain and refer the account for review
    end if
end for
Compute classification and selective-prediction metrics
Compare baselines and ablations under each evaluation protocol
Report statistical uncertainty and calibration assumptions
```

hyperparameter selection, will be fitted using training data only. Events without valid timestamps will be excluded from timestamp-dependent calculations. Duplicate records will be removed using a documented rule. For the primary inductive group-disjoint protocol, groups will not overlap between training, validation, calibration, and test partitions, and cross-partition co-activity edges will be excluded. Test labels will not be used for feature construction, graph construction, model selection, or calibration. Split manifests and preprocessing settings will be retained for reproducibility.

### 6.2 Evaluation Protocols

- **P0: Random account-level split.** Accounts are randomly partitioned into training, validation, calibration (if required), and test sets. Related groups may occur in multiple partitions; this is a reference protocol, not evidence of unseen-campaign generalization.
- **P1: Group-disjoint evaluation.** Each annotated campaign or declared proxy group is assigned to only one partition. The term campaign-disjoint is reserved for independently annotated campaigns.
- **P2: Chronological evaluation.** Training and evaluation are separated by a documented time cutoff, and features use only information available at prediction time. Label-availability time is recorded separately to distinguish temporal shift from label staleness.

- **P3: Cross-dataset evaluation.** A model trained on one corpus is evaluated on a separately collected corpus when label definitions and features can be mapped consistently. Direct transfer and any target-side adaptation are reported as separate settings.

These protocols measure different types of generalization and will not be interpreted as interchangeable.

### 6.3 Baselines and Model Variants

The proposed model will be compared, where compatible implementations and inputs are available, with the following baselines:

1. **Traditional classifiers:** logistic regression, support vector machine, and random forest using profile features.
2. **Activity-based classifier:** profile and activity-rhythm features without relational coordination information.
3. **Graph baselines:** GraphSAGE and graph attention networks supplied with equivalent permitted graph information.
4. **Temporal graph baselines:** TGN or TGAT when the dataset contains the event representation required by the implementation.
5. **Coordination baseline:** shared-object co-activity graph without null-model correction.
6. **Shift-robust baselines:** group-DRO, group-aware reweighting, and an IRM-style method when the group structure and implementation support a fair comparison.

Methods that require unavailable fields or cannot be implemented faithfully will be reported as inapplicable rather than silently replaced. Baselines will use identical partitions and comparable feature availability wherever possible.

### 6.4 Experiment E1: Random-Split versus Group-Disjoint Performance

**Objective.** E1 tests whether random account-level evaluation gives different performance estimates from evaluation on held-out campaigns or proxy groups.

**Procedure.** Each eligible model will be evaluated under P0 and P1 using the same feature definitions and model-selection rules. Hyperparameters will be selected without using test labels. The numbers of accounts, groups, class proportions, and excluded records will be reported for each partition.

**Metrics.** Macro-F1, PR-AUC, ROC-AUC, and Matthews correlation coefficient (MCC) will be reported when defined. The observed difference for metric $M$ is

$$\Delta M = M_{\mathrm{P0}} - M_{\mathrm{P1}}. \tag{29}$$

A positive value indicates a higher measured score under P0, but does not by itself establish that campaign overlap caused the difference.

**Single-class groups.** If a held-out group contains only one class, metrics requiring both classes or both prediction outcomes may be undefined. Such values will be marked as not estimable, not replaced by favorable values. The number of affected groups and their class distributions will be reported. Pooled out-of-fold predictions may support an overall score when appropriate, but will not substitute for per-group reporting.

**Evidence sought.** E1 quantifies the observed P0–P1 difference and its uncertainty; the magnitude and direction must be determined empirically.

### 6.5 Experiment E2: Contribution of Temporal Coordination Modeling

**Objective.** E2 tests whether temporal coordination adds predictive information beyond profile characteristics and individual activity rhythms under group-disjoint evaluation.

**Compared variants.** The experiment compares (i) profile-only, (ii) profile plus activity rhythm, (iii) profile plus temporal coordination, and (iv) the complete model combining all three evidence sources.

**Procedure and metrics.** Variants will use the same group-disjoint splits, training conditions, and model-selection procedure. The principal metrics are Macro-F1, PR-AUC, ROC-AUC, and MCC where defined. The incremental contribution of coordination is assessed by comparing the complete model with the profile-plus-rhythm model. The contribution of the rhythm representation is assessed by comparing the profile-plus-rhythm model with the profile-only model. No gain will be claimed unless supported by measured results and appropriate uncertainty estimates.

### 6.6 Experiment E3: Calibration and Group-Level Prediction-Set Reliability

**Objective.** E3 assesses empirical prediction-set coverage and campaign- or group-level miscoverage under the implemented calibration procedures.

**Compared methods.** Raw classifier probabilities, temperature scaling, and supported prediction-set calibration methods will be compared when their data requirements are met. Account-level and group-blocked calibration will be compared only with their assumptions and implementation requirements stated explicitly.

**Procedure and metrics.** The prediction model will be fitted without calibration or test labels. Calibration thresholds will use the designated calibration partition, and the test partition will be used only for final evaluation. For group $g$, miscoverage is

$$L_g = \frac{1}{|\mathcal{U}_g|} \sum_{u \in \mathcal{U}_g} \mathbb{I}\left[y_u \notin \Gamma(\mathbf{x}_u)\right]. \tag{30}$$

The study will report overall empirical coverage, group-level miscoverage, the number and size of independent calibration and test groups, calibration sample size, and class-conditional coverage where sample sizes permit. The impact of small numbers of independent groups on attainable risk levels and uncertainty will be stated.

**Guarantee interpretation.** Coverage and group miscoverage are empirical quantities. A formal expected-risk guarantee will be stated only if the implemented algorithm, fixed-predictor requirements, sampling assumptions, and finite-sample correction justify it. Group-disjoint partitioning alone does not establish exchangeability between calibration groups and future groups. No guarantee under arbitrary campaign shift or for every individual group is inferred from empirical coverage.

### 6.7 Experiment E4: Ablation of Null-Model Correction and Graph Construction

**Objective.** E4 isolates the effects of temporal weighting, null-model correction, and graph sparsification.

**Compared graph variants.** The variants are (i) no coordination graph, (ii) shared-object graph without temporal weighting, (iii) temporally weighted graph without null correction, (iv) temporally weighted graph with null correction, and (v) null-corrected graph with minimum-support filtering and sparsification.

**Procedure.** The classifier, partitions, and training settings will be held constant wherever possible. The null model will account for object-level activity and coarse temporal variation to the extent supported by the dataset. Randomization settings will be documented before final testing. Where suitable labeled examples exist, legitimate synchronized activity will be examined because shared external events can produce synchronous behavior among benign accounts.

**Metrics.** Detection metrics include Macro-F1, PR-AUC, ROC-AUC, and MCC where defined. Graph diagnostics include edge count, degree statistics, connected components, and the fraction of edges satisfying the minimum-support rule. Graph-construction time and peak memory will also be measured. A normalized co-activity score will not be called a significance test unless a valid testing procedure is implemented.

### 6.8 Experiment E5: Selective Prediction and Automatic-Labeling Trade-off

**Objective.** E5 measures the relationship between prediction-set reliability, automatic labeling, and human-review workload.

**Procedure.** The selected prediction-set method will be evaluated at predefined thresholds. An automatic label is assigned only when the set contains exactly one class. Empty and non-singleton sets are treated as abstentions. The study reports empirical coverage, singleton rate, abstention rate, selective accuracy, and precision, recall, and F1 among automatically labeled accounts when defined. Selective accuracy is undefined if no account receives an automatic label.

**Analysis.** Selective accuracy will be plotted against singleton rate, and prediction-set coverage will be reported separately. Coverage does not guarantee correctness among the subset selected for automatic labeling. A guarantee for marginal set miscoverage will not be interpreted as a selective-risk, precision, or false-discovery-rate guarantee.

### 6.9 Experiment E6: Robustness to Incomplete Activity and External Synchronization

**Objective.** E6 evaluates sensitivity to missing events, timestamp perturbations, low activity, and legitimate synchronized behavior.

**Perturbations.** The experiment includes (i) removal of a predefined fraction of events, (ii) timestamp noise within documented bounds, (iii) separate evaluation of low-activity accounts or groups, and (iv) evaluation on labeled legitimate synchronized activity where reliable examples exist. Perturbation levels will be fixed before final testing and applied consistently to the proposed method and relevant baselines. The unperturbed data provide the reference condition.

**Metrics and analysis.** Macro-F1, PR-AUC, false-positive rate, group-level miscoverage, singleton rate, and computational cost will be measured as applicable. Changes relative to the unperturbed condition will be reported. Missing-observation sensitivity and false positives due to legitimate synchronization will be analyzed as distinct failure modes.

### 6.10 Experiment E7: Computational Scalability

**Objective.** E7 measures the computational overhead of temporal graph construction, null estimation, graph learning, and calibration.

**Procedure.** Progressively larger dataset subsets will be evaluated subject to available memory and runtime. For each subset, the numbers of accounts, events, candidate edges, and retained edges will be recorded. Measurements include graph-construction time, null-model estimation time, training time, per-account and total inference time, peak memory, and graph size. Hardware, software versions, accelerator use, and relevant settings will be documented.

**Analysis.** Runtime and memory will be plotted against account, event, and edge counts. The analysis will identify whether graph construction or inference dominates total cost. Complexity claims will be supported by the implemented procedure and measured behavior.

### 6.11 Experiment E8: Cross-Dataset Generalization

**Objective.** E8 evaluates transfer to an independently collected dataset with potentially different collection conditions, populations, and annotation procedures.

**Procedure.** One dataset will be used for training and validation, and a second compatible dataset will be reserved for external testing. Label mapping will be documented before evaluation. Features unavailable in the target dataset will not be silently reconstructed. The primary transfer setting will not use target test labels for model selection or threshold optimization. Any target-side adaptation will be reported separately.

**Metrics and interpretation.** Macro-F1, PR-AUC, ROC-AUC, MCC, false-positive rate, and supported prediction-set and selective-prediction metrics will be reported. Differences will be interpreted in light of label semantics, feature compatibility, class proportions, and collection conditions; they will not automatically be attributed solely to campaign shift.

Table 2: Summary of the experimental plan and principal evaluation criteria.

| ID | Experiment name | Primary comparison | Principal measures |
|---|---|---|---|
| E1 | Random versus group-disjoint evaluation | P0 versus P1 | Macro-F1, PR-AUC, ROC-AUC, MCC |
| E2 | Temporal coordination contribution | Profile, rhythm, and coordination variants | Detection metrics and incremental performance |
| E3 | Calibration and group-level reliability | Supported calibration procedures | Coverage, group miscoverage, calibration-unit count |
| E4 | Null-model and graph ablation | Uncorrected versus corrected graph variants | Detection metrics, graph diagnostics, construction cost |
| E5 | Selective-prediction trade-off | Prediction-set thresholds | Selective accuracy, singleton rate, abstention, coverage |
| E6 | Robustness evaluation | Perturbed versus unperturbed observations | Detection degradation, false-positive rate, reliability |
| E7 | Computational scalability | Increasing dataset and graph sizes | Runtime, memory, graph size, inference cost |
| E8 | Cross-dataset generalization | Within-dataset versus external transfer | Detection, calibration, and selective metrics |

### 6.12 Statistical Analysis and Reporting

Each experiment will report both account counts and independent group counts. Repeated random seeds measure training variability and will not be treated as independent campaigns. When methods are evaluated on the same groups, paired comparisons will be used. Confidence intervals will be estimated by resampling at the independent-group level when the number of groups supports meaningful estimation. A paired permutation test or Wilcoxon signed-rank test may be used for suitable paired group-level measurements; the selected test and its assumptions will be stated.

A confirmatory family of hypotheses and primary metrics will be specified before final testing. Holm correction will be applied within the predefined family of confirmatory tests. Exploratory analyses will be labeled as such. If there are too few independent groups for reliable inference, results will be reported descriptively and the limitation stated. Minimum practically meaningful effects must be chosen before examining final test outcomes rather than selected to match observed results.

### 6.13 Summary of Experiments

Table 2 summarizes the eight experiments.

Together, these experiments assess whether temporal coordination contributes information beyond account-level features, whether this contribution persists under group-disjoint evaluation, and how prediction reliability and computational cost change under demanding conditions. Conclusions will be based on executed experiments, audited data properties, and reported uncertainty rather than on the protocol alone.

## 7 Conclusion

This paper presents a campaign-shift-aware framework for social bot detection that integrates temporal coordination modeling, null-model correction, graph neural networks, and uncertainty-aware selective prediction. The proposed evaluation methodology examines generalization to unseen campaigns, prediction reliability, and the trade-off between automatic classification

and human review. Campaign-disjoint, chronological, and cross-dataset experiments, together with controlled ablations, are required to establish the contribution of each component. The present manuscript defines a testable methodology; it does not claim empirical superiority before the experiments are executed. Campaign-level risk control is treated as empirical unless the calibration-unit assumptions and finite-sample requirements are verified. The next step is to execute the P0–P1 comparison and core ablations on audited datasets, report independent-group counts and uncertainty, and evaluate robustness and scalability.

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